

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	11 June 2026
Team ID	
Project Name	ShopEZ: E-Commerce Application
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my name, email, password, and confirming my password.	2	High	Developer
Sprint-1	Registration	USN-2	As a user, I can see validation messages if required registration fields are empty or invalid.	1	High	Developer
Sprint-1	Login	USN-3	As a user, I can log into the application by entering my email and password.	2	High	Developer
Sprint-1	Login	USN-4	As a user, I can see an error message when invalid login credentials are entered.	1	High	Developer

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Authentication	USN-5	As a user, I can stay logged in securely using authentication tokens.	3	High	Developer
Sprint-1	Dashboard	USN-6	As a user, I can view the homepage/dashboard after login.	2	High	Developer
Sprint-1	Database	USN-7	As a developer, I can connect the backend server to MongoDB database.	3	High	Developer
Sprint-1	API Setup	USN-8	As a developer, I can create basic API routes for users, products, stores, and orders.	4	High	Developer
Sprint-1	UI Setup	USN-9	As a developer, I can create the basic frontend layout with navigation.	2	Medium	Developer

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	01 June 2026	06 June 2026	20	06 June 2026
Sprint-2	20	6 Days	08 June 2026	13 June 2026	18	14 June 2026
Sprint-3	20	6 Days	15 June 2026	20 June 2026	19	21 June 2026
Sprint-4	20	6 Days	22 June 2026	27 June 2026	20	27 June 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Since this is a solo project, velocity represents how much work one developer completes in one sprint.

Sprint Duration = 6 days

Planned Story Points = 20

Average Velocity per day = $20 / 6$

Average Velocity per day = 3.33 story points per day

Average velocity across all sprints:

Average Velocity = $(20 + 18 + 19 + 20) / 4$

Average Velocity = $77 / 4$

Average Velocity = 19.25 story points per sprint

So, for this solo project, the average velocity is:

19.25 story points per sprint

Burndown Chart:

In the ShopEZ e-commerce application, the burndown chart is used to monitor the remaining story points during each sprint. Since this is a solo project, the chart helps the developer track whether the planned work is being completed on time.

The vertical axis represents the remaining story points, and the horizontal axis represents the sprint days. As development progresses, the remaining work should reduce gradually and reach zero by the end of the sprint.

Sprint-1 Burndown Chart Data

Day	Ideal Remaining Story Points	Actual Remaining Story Points
Day 0	20	20
Day 1	16.67	17
Day 2	13.34	14
Day 3	10.01	10
Day 4	6.68	7
Day 5	3.35	3

Day	Ideal Remaining Story Points	Actual Remaining Story Points
Day 6	0	0

Explanation

The ideal remaining story points show the planned progress of the sprint. The actual remaining story points show the real progress made during development.

For Sprint-1, the total work was 20 story points and the duration was 6 days. By the end of Day 6, all story points were completed. Therefore, the sprint was completed successfully within the planned time.